

Putnam 2014

A1 Prove that every nonzero coefficient of the Taylor series of $(1 - x + x^2)e^x$ about $x = 0$ is a rational number whose numerator (in lowest terms) is either 1 or a prime number.

Solution. We consider the expansion $(1 - x + x^2) \sum_{i=0}^{\infty} \frac{1}{i!} x^i$. The coefficient of the constant term is 1 and the coefficient of the x -term is $\frac{1}{2} - 1 = -\frac{1}{2}$. The coefficient of x^{k+1} for all $k \geq 1$ is given by $\frac{1}{(k+1)!} - \frac{1}{k!} + \frac{1}{(k-1)!} = \frac{1 - (k+1) + k(k+1)}{(k+1)!} = \frac{k^2}{(k+1)!} = \frac{k}{(k-1)!(k+1)}$. If k is prime we are done. Now assume that it's not. If k is not a prime power, write $k = ab$ with $1 < a, b < k$ and $\gcd(a, b) = 1$ (for example, let p to be a prime divisor of k and let r to be the maximum power of p dividing k . Then since k is not a prime power, $p^r < k$ and $k/(p^r)$ is relatively prime to p^r by the maximality of r). Since $a, b < k$, $a|(k-1)!$ and $b|(k-1)!$ and with $\gcd(a, b) = 1$, this implies that $k = ab|(k-1)!$. Otherwise, $k = p^r$ for some prime p and $r \geq 2$. Using the formula $v_p((r)!) = \sum_{i=1}^{\infty} \lfloor \frac{r}{p^i} \rfloor$, we have $v_p((k-1)!) = \sum_{i=1}^{\infty} \lfloor \frac{k-1}{p^i} \rfloor = \sum_{i=1}^{\infty} \lfloor \frac{p^r-1}{p^i} \rfloor = p^{r-1} - 1 + p^{r-2} - 1 + \cdots + (p-1) \geq 1 + 1 + \cdots + 1 = r-1$ since $p \geq 2$. Thus $p^{r-1} | (k-1)!$ and so when taking the lowest term the numerator can either be 1 or p .

A2 Let A be the $n \times n$ matrix whose entry in the i -th row and j -th column is

$$\frac{1}{\min(i, j)}$$

for $1 \leq i, j \leq n$. Compute $\det(A)$.

Answer. $(-1)^n \frac{1}{n![(n-1)!]^2}$

Solution. We use the well-known matrix identity that row reduction preserves determinant, and we will do row reduce profusely. For brevity, we will denote $f(i) = \frac{1}{i}$ for all $i \geq 1$. Denoting a_{ij} as the i -th row and the j -th column. Then we have $a_{ij} = f(\min(i, j))$. Now, for each iteration stepped i , denoting row i as r_i and we will do $r_j := r_j - r_i$ for all $j \geq i$. We show that after the k -th iteration below would be the value for a_{ij} :

- For $i = 1$, we have $a_{ij} = f(1)$ as always.
- For $i \leq k$, we have $a_{ij} = 0$ for all $j < i$, and $a_{ij} = f(i) - f(i-1)$ for all $j \geq i$.
- For $i > k$, $a_{ij} = 0$ for $j \leq k$, and $a_{ij} = f(\min(i, j)) - f(k)$ otherwise.

To prove this by inducting on k , the base case is given when all the numbers after the first row are subtracted by the corresponding number in the first row, so for $i > 1$, a_{ij} becomes $a_{ij} = f(\min(i, j)) - f(1)$, and the condition above is satisfied. Suppose that the conjecture holds after k -th step for some k . At $k+1$ -th step, all rows after the $k+1$ -th row is subtracted against the corresponding index in $k+1$ -th row. The $k+1$ -th row is given by the following:

$$(0 \quad \cdots 0 \quad f(k+1) - f(k) \quad \cdots f(k+1) - f(k))$$

where the first k entries are 0. Now after the $k+1$ -th iteration, for all $i > k+1$, if $j \leq k$ then a_{ij} becomes $0 - 0 = 0$ and if $j > k$ we have a_{ij} becomes $(f(\min(i, j)) - f(k)) - f(k+1) - f(k) = f(\min(i, j)) - f(k+1)$. This entry is 0 if $j = k+1$ since $i > k+1$. For all $i \leq k$ the rows are unaffected by this row reduction, so we still have $a_{ij} = 0$ for all $j < i$, and $a_{ij} = f(i) - f(i-1)$ for all $j \geq i$. Thus the claim is proven.

To finish the proof, after $n-1$ iterations, we have, for all $j < i$, $a_{ij} = 0$. Thus A is no upper triangular, and the determinant is simply the product of the diagonal entries. We also

have $a_{ii} = 1$ for $i = 1$ and $f(i) - f(i-1)$ for $i \geq 2$. Now $f(i) - f(i-1) = \frac{1}{i} - \frac{1}{i-1} = -\frac{1}{i(i-1)}$. Hence we have

$$\det(A) = \prod_{i=2}^n -\frac{1}{i(i-1)} = -(-1)^{n-1} \frac{1}{n[(n-1)!]^2}$$

A3 Let $a_0 = 5/2$ and $a_k = a_{k-1}^2 - 2$ for $k \geq 1$. Compute

$$\prod_{k=0}^{\infty} \left(1 - \frac{1}{a_k}\right)$$

in closed form.

Answer. $\frac{3}{7}$.

Solution. Consider, in general, $a_0 = a + \frac{1}{a}$ for some $a > 1$, then we show that the answer is $\frac{a^2 - 1}{a^2 + a + 1}$. Here the problem is a special case with $a = 2$.

Claim. $a_k = a^{2^k} + \frac{1}{a^{2^k}}$.

Indeed, this holds for a_0 , and using induction we have

$$\left(a^{2^{k-1}} + \frac{1}{a^{2^{k-1}}}\right)^2 - 2 = a^{2^k} + \frac{1}{a^{2^k}} + 2 - 2 = a^{2^k} + \frac{1}{a^{2^k}}$$

Thus our term now becomes

$$\prod_{k=0}^{\infty} \frac{a^{2^{k+1}} - a^{2^k} + 1}{a^{2^{k+1}} + 1} = \prod_{k=0}^{\infty} \frac{1 - a^{-2^k} + a^{-2^{k+1}}}{1 + a^{-2^{k+1}}}$$

Let's evaluate the limits of numerator and denominator separately. For the denominator we simply have

$$\prod_{k=0}^n (1 + a^{-2^{k+1}}) = (1 + a^{-2})(1 + a^{-4}) \cdots (1 + a^{-2^{n+1}}) = \sum_{i=0}^{2^{n+1}-1} a^{-2i}$$

(well the right equality is easy to verify). Taking $n \rightarrow \infty$ we have

$$\sum_{i=0}^{2^{n+1}-1} a^{-2i} = (1 - a^{-2})^{-1}$$

For the numerator, we claim that

$$\prod_{k=0}^n (1 - a^{-2^k} + a^{-2^{k+1}}) = \left(\sum_{i=0}^{2^{n+1}-2} r_i (a^{-i} + a^{-(2^{n+2}-2-i)}) \right) + (-1)^{n-1} a^{-(2^{n+1}-1)} \quad r_i = \begin{cases} 1 & 3 \mid i \\ -1 & 3 \mid i-1 \\ 0 & 3 \mid i-2 \end{cases}$$

Notice also that $r_{2^{n+1}-1} = (-1)^{n-1}$. Again we use induction. For base case we just have $1 - a^{-1} + a^{-2}$, and the RHS is just $1(1 + a^{-2}) - a^{-1} = 1 - a^{-1} + a^{-2}$ (and therefore it matches). For induction step, let's consider

$$\left(\left(\sum_{i=0}^{2^{n+1}-2} r_i (a^{-i} + a^{-(2^{n+2}-2-i)}) \right) + (-1)^{n-1} a^{-(2^{n+1}-1)} \right) (1 - a^{-2^{n+1}} + a^{-2^{n+2}})$$

Now we need to consider the following:

- $0 \leq i < 2^{-(n+1)}$, then naturally we have the coefficient as r_i .
- $2^{-(n+1)} \leq i \leq 2^{n+2} - 1$, then we have the coefficient as $r_{2^{n+2}-2-i} - r_{i-2^{n+1}}$. We see that r is a mod 3 function, so we can consider $2^{n+1} \equiv 1$ or $2^{n+2} \equiv 2$. For the first case we have $r_{-i} + r_{i-1}$, and for $i = 0, 1, 2$ this gives

$$r_0 - r_{-1} = 1 = r_0; \quad r_{-1} - r_0 = -1 = r_1; \quad r_{-2} - r_{-1} = 0 = r_2$$

and for the second case we have $r_{2-i} - r_{i-2}$, which gives

$$r_2 - r_{-2} = 1 = r_0; \quad r_1 - r_{-1} = -1 = r_1; \quad r_0 - r_0 = 0 = r_2$$

which means that the coefficient is just going to be r_i .

- The case $i \geq 2^{n+2}$ follows from that the coefficient is symmetric w.r.t. $2^{n+2} - 1$ at $n + 1$.

Thus as $n \rightarrow \infty$, this should behave like

$$1 - a^{-1} + a^{-3} - a^{-4} + \dots = (1 - a^{-1})(1 + a^{-3} + a^{-6} + \dots) = (1 - a^{-1})(1 - a^{-3})^{-1} = \frac{1}{1 + a^{-1} + a^{-2}}$$

(Well the latter half of the terms, i.e. $i \geq 2^{n+1} - 2$, don't quite follow this rule, but the effect is $\rightarrow 0$ since $a^{-2^{n+1}} + \dots + a^{-2^{n+2}} \leq a^{-2^{n+1}}(1 - a^{-1})^{-1} \rightarrow 0$). Thus the limit we're looking for is now

$$\frac{1 - a^{-2}}{a + a^{-1} + a^{-2}} = \frac{a^2 - 1}{a^2 + a + 1}$$

as desired.

A4 Suppose X is a random variable that takes on only nonnegative integer values, with $\mathbb{E}[X] = 1$, $\mathbb{E}[X^2] = 2$, and $\mathbb{E}[X^3] = 5$. (Here $\mathbb{E}[Y]$ denotes the expectation of the random variable Y .) Determine the smallest possible value of the probability of the event $X = 0$.

Answer. $\frac{1}{3}$. This is achieved when X only takes values 0, 1, 3 with probabilities $\frac{1}{3}, \frac{1}{2}, \frac{1}{6}$, respectively.

Solution.

Let $p_i = \mathbb{E}[X = i]$ for $i = 0, 1, 2$, and $p_3 = \mathbb{E}[X \geq 3]$. Consider, now, the random variable Y , taking only nonnegative integer values, such that $\mathbb{E}[Y = y] = \frac{\mathbb{E}[X=y+3]}{p_3}$, and $b_i = \mathbb{E}[Y^i]$ for $i = 1, 2, 3$. Then:

$$\mathbb{E}[X^i] = p_0 0^i + p_1 1^i + p_2 2^i + p_3 \mathbb{E}[(Y + 3)^i]$$

for $i = 1, 2, 3$. Notice also

$$\mathbb{E}[(Y + 3)] = b_1 + 3 \quad \mathbb{E}[(Y + 3)^2] = \mathbb{E}[Y^2] + 6\mathbb{E}[Y] + 9 = b_2 + 6b_1 + 9$$

$$\mathbb{E}[(Y + 3)^3] = \mathbb{E}[Y^3] + 9\mathbb{E}[Y^2] + 27\mathbb{E}[Y] + 27 = b_3 + 9b_2 + 27b_1 + 27$$

This gives us the following:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & b_1 + 3 \\ 0 & 1 & 4 & b_2 + 6b_1 + 9 \\ 0 & 1 & 8 & b_3 + 9b_2 + 27b_1 + 27 \end{pmatrix} \begin{pmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 5 \end{pmatrix}$$

We note that b_1, b_2, b_3 are all nonnegative by the definition of Y , so the left matrix is invertible, and gives the following solution:

$$p_3 = \frac{1}{D} \quad p_2 = \frac{b_3 + 5b_2 + 6b_1}{2D} \quad p_1 = \frac{b_2 + 4b_1 + 3}{D} \quad p_0 = \frac{b_3 + 5b_2 + 8b_1 + 4}{2D}$$

where $D = b_3 + 6b_2 + 11b_1 + 6$. Therefore our required quantity is now

$$\frac{b_3 + 5b_2 + 8b_1 + 4}{2(b_3 + 6b_2 + 11b_1 + 6)} = \frac{1}{3} + \frac{b_3 + 3b_2 + 2b_1}{6(b_3 + 6b_2 + 11b_1 + 6)} \geq \frac{1}{3}$$

since $b_1, b_2, b_3 \geq 0$.

Remark. Here's a more elegant solution that I found after. Now notice that $\mathbb{E}[(X-1)^k] = 0, 1, 1$ for $k = 1, 2, 3$. Denote $p_m = \mathbb{P}[X = m]$. We leverage the fact that $(X-1)^3 - (X-1) = 0$ for $X = 0, 1, 2$, and also $(X-1)^2 - (X-1) = 0$ for $X = 0, 1, 2$, to get

$$1 = \sum_{m \geq 3} p_m [(X-1)^3 - (X-1)] = 2p_0 + \sum_{m \geq 3} p_m [(X-1)^2 - (X-1)]$$

On the other hand we have

$$\sum_{m \geq 3} p_m [(X-1)^2 - (X-1)] \leq \max_{m \geq 3} \frac{(X-1)^2 - (X-1)}{(X-1)^3 - (X-1)} = \max_{m \geq 3} \frac{1}{X} = \frac{1}{3}$$

which forces $p_0 \geq \frac{1}{3}$. Equality can be found as per the first solution.

- A6** Let n be a positive integer. What is the largest k for which there exist $n \times n$ matrices $M_1, \dots, M_k$ and $N_1, \dots, N_k$ with real entries such that for all i and j , the matrix product $M_i N_j$ has a zero entry somewhere on its diagonal if and only if $i \neq j$?

Answer. n^n .

Solution. We consider a more general setting where M_i 's are $n \times m$ and N_j 's are $m \times n$, in which we show the answer is m^n . Equality can be taken via $N_i = M_i^T$, and also the n rows of M_i are $e_{i_1}, e_{i_2}, \dots, e_{i_n}$ where $(i_1 i_2 \dots i_n)$ are the base- m expansion of $i-1$, $i = 1, \dots, k = m^n$.

To show the converse, we first note the following:

Lemma 1. Let A and B be two $n \times n$ matrices, and denote $\circ$ the Hadamard product (i.e. elementwise multiplication). Then the $\text{rank}(A \circ B) \leq \text{rank}(A) \cdot \text{rank}(B)$.

Proof. If one of A and B is zero matrix then the product is a zero matrix (hence rank 0). Otherwise, if $\text{rank}(A) = m_1$ and $\text{rank}(B) = m_2$, then columns of A are in $\text{span}\{u_1, \dots, u_{m_1}\}$ and columns of B are in $\text{span}\{v_1, \dots, v_{m_2}\}$. Therefore $A \circ B$ has columns in $\text{span}\{u_i \cdot v_j : i = 1, \dots, m_1, j = 1, \dots, m_2\}$, so its rank is upper-bounded by $m_1 \cdot m_2$, as desired. $\square$

Now, for each $p = 1, \dots, n$, let A_p be the $k \times m$ matrix determined by stacking the p -th row of $(M_1, \dots, M_k)$, and B_p be the $m \times k$ matrix determined by stacking the p -th column of $(M_1, \dots, M_k)$. Denote $C_p = A_p B_p$. Note that $\text{rank}(C_p) \leq m$. Now, each C_p has nonzero diagonal entries, but for each (i, j) with $i \neq j$, one of C_p 's ($p \in \{1, \dots, n\}$) has zero entry at (i, j) . Therefore, the following matrix

$$C = C_1 \circ C_2 \circ \dots \circ C_n$$

is a $k \times k$ diagonal matrix with nonzero diagonal entries, and is of full rank k . We now finish with the lemma via the following:

$$k = \text{rank}(C) \leq \prod_{p=1}^n \text{rank}(C_p) \leq \prod_{p=1}^n m = m^n$$

as desired.

- B1** A base 10 over-expansion of a positive integer N is an expression of the form $N = d_k 10^k + d_{k-1} 10^{k-1} + \cdots + d_0 10^0$ with $d_k \neq 0$ and $d_i \in \{0, 1, 2, \dots, 10\}$ for all i . For instance, the integer $N = 10$ has two base 10 over-expansions: $10 = 10 \cdot 10^0$ and the usual base 10 expansion $10 = 1 \cdot 10^1 + 0 \cdot 10^0$. Which positive integers have a unique base 10 over-expansion?

Answer. All positive integers without any 0 in their decimal expansion.

Solution. We use the fact that every positive integer has a unique base-10 expansion (that is, all digits $0, \dots, 9$). Therefore, a non-unique over expansion is equivalent to the existence of an over expansion with the ‘digit’ 10 being used.

Consider the expansion $n = \sum_{i=0}^k d_i 10^i$ with $0 \leq d_i \leq 9$ and $d_k \neq 0$. If n has 0 as one of the digits, then there exists a position $j > 0$ such that $d_j > 0$ but $d_{j-1} = 0$. Then we can replace d_j with $d_j - 1$ and d_{j-1} with 10, giving two over-expansions here.

Next we show that any number n with $d_i = 10$ for some i in its over-expansion must contain a 0 somewhere in its decimal expansion. Indeed, let j be the minimal index with

$$d_j = 10. \text{ Then } n \equiv \sum_{i=0}^j d_i 10^i \equiv \sum_{i=0}^{j-1} d_i 10^i \pmod{10^{j+1}}. \text{ We see that } 0 \leq \sum_{i=0}^{j-1} d_i 10^i < 10^j$$

by the minimality of j , and with $d_j = 10, n \geq 10^{j+1}$. Thus this implies that the digit at position j is indeed 0.

- B2** Suppose that f is a function on the interval $[1, 3]$ such that $-1 \leq f(x) \leq 1$ for all x and $\int_1^3 f(x) dx = 0$. How large can $\int_1^3 \frac{f(x)}{x} dx$ be?

Answer. $\ln \frac{4}{3}$.

Solution. Equality can be attained by taking $f(j) = 1$ for all $1 \leq j < 2$ and $f(j) = -1$ for all $2 \leq j \leq 3$. We show that this is the maximum by the following: if $g(x)$ is defined as $\int_1^x f(y) dy$, we have $g(1) = g(3) = 0$. Also since $f(x) \in [-1, 1]$ for all $x \in [1, 3]$, and by Mean value theorem, we have, for every x in the said interval, $g'(c) = f(c) = \frac{g(x) - g(1)}{x - 1}$ for some constant c in the interval $(1, x)$, so $|\frac{g(x)}{x-1}| \leq 1$. Similarly $|\frac{g(x)}{x-3}| \leq 1$. This means that $g(x) \leq x - 1$ and $g(x) \leq 3 - x$ must hold simultaneously. Using this fact and integrating by parts give:

$$\begin{aligned} \int_1^3 \frac{f(x)}{x} dx &= \frac{g(x)}{x} \Big|_1^3 + \int_1^3 \frac{g(x)}{x^2} dx \\ &= (0 - 0) + \int_1^3 \frac{g(x)}{x^2} dx \\ &\leq \int_1^2 \frac{x-1}{x^2} dx + \int_2^3 \frac{3-x}{x^2} dx \\ &= [\ln x + \frac{1}{x}]_1^2 + [-\frac{3}{x} - \ln x]_2^3 \\ &= \ln 2 - \ln 1 + \frac{1}{2} - 1 + \frac{3}{2} - 1 - \ln 3 + \ln 2 \\ &= \ln \frac{4}{3} \end{aligned}$$

as desired.

- B3** Let A be an $m \times n$ matrix with rational entries. Suppose that there are at least $m + n$ distinct prime numbers among the absolute values of the entries of A . Show that the rank of A is at least 2.

Solution. By the theorem of unique prime factorization, if p, q, r, s are prime numbers with $pq = rs$ then $p = r, q = s$ or $p = s, q = r$ (so the four numbers cannot be pairwise

distinct). The fact that there's at least one prime (and hence nonzero) number in A implies that the rank of A cannot be zero, so we can now assume that the rank of A is 1, which is equivalent to assuming that there exists rational numbers $a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n$ such that $A_{ij} = a_i b_j$.

Now consider a graph (V, E) with n vertices. and we consider adding coloured edge by the following mechanism: for a row i , if $x_1 < x_2 < \dots < x_k$ are the all the indices such that A_{ix_j} are among the $m+n$ distinct prime numbers, then we add an edge coloured i between x_j and x_{j+1} for each $1 \leq j \leq k-1$. This means if row i has i_k prime numbers the there will be $i_k - 1$ edges coloured i . Our colouring also ensures that there will be no monochromatic cycle in our graph, and there are at least $\sum_{k=1}^m (i_k - 1) = (\sum_{k=1}^m i_k) - m = (m + n - m) = n$.

We first see what happens if there are two vertices c_1, c_2 with two edges coloured k_1 and k_2 . This means $A_{k_1 c_j} = a_{k_1} b_{c_j}$ are all prime numbers for all combinations of $i \in \{1, 2\}$ and $j \in \{1, 2\}$. Notice also that $A_{k_1 c_1} A_{k_2 c_2} = a_{k_1} b_{c_1} a_{k_2} b_{c_2} = a_{k_1} b_{c_2} a_{k_2} b_{c_1} = A_{k_1 c_2} A_{k_2 c_1}$, contradicting that the four prime numbers must be pairwise distinct.

Hence we know that there is at most an edge between two vertices, and since there are exactly n vertices and at least n edges, there exists a cycle comprising at least two different colours (since we have proven that there cannot be a monochromatic cycle above). Let $x_1, x_2, \dots, x_k$ to be the cycle, with $x_i x_{i+1}$ connected by colour r_i for each $1 \leq i \leq k$. For each i , $A_{r_i x_i}$ and $A_{r_i x_{i+1}}$ are both primes, and let $p_{r_i x_i}, p_{r_i x_{i+1}}$ be the primes. Now $\frac{p_{r_i x_i}}{p_{r_i x_{i+1}}} = \frac{A_{x_i i}}{A_{x_i(i+1)}} = \frac{a_{x_i} b_i}{a_{x_i} b_{i+1}} = \frac{b_i}{b_{i+1}}$ (the fact that both entries are prime, i.e. nonzero, means that we don't have to worry about the validity of division). Thus we have

$$1 = \prod_{i=1}^k \frac{b_i}{b_{i+1}} = \prod_{i=1}^k \frac{p_{r_i x_i}}{p_{r_i x_{i+1}}}$$

and by the theorem of unique prime factorization, $\prod_{i=1}^k p_{ii}$ and $\prod_{i=1}^k p_{(i+1)i}$ also implies that $\{p_{r_i x_i} : 1 \leq i \leq k\} = \{p_{r_i x_{i+1}} : 1 \leq i \leq k\}$. Since $p_{r_i x_i}$ corresponds to the entry (r_i, x_i) and $p_{r_i x_{i+1}}$ the entry (r_i, x_{i+1}) , and each $x_1, x_2, \dots, x_k$ assumed to be distinct and each of the $m+n$ primes are distinct, we have $p_{r_i x_{i+1}} = p_{r_{i-1} x_i}$, $r_i = r_{i-1}$, so $r_1 = r_2 = \dots = r_k$. This also means that the only possibility is all edges of the cycle coloured the same colour r_1 , contradiction.

B4 Show that for each positive integer n , all the roots of the polynomial

$$\sum_{k=0}^n 2^{k(n-k)} x^k$$

are real numbers.

Solution. Let $f(x)$ be the polynomial, which obviously takes positive values when $x \geq 0$. Consider, now, all x 's with $x = -(2^m)$. Then

$$f(x) = \sum_{k=0}^n 2^{k(n-k)} x^k = \sum_{k=0}^n 2^{k(n-k)} (-2^m)^k = \sum_{k=0}^n (-1)^k 2^{k(n+m-k)}$$

We first notice that when k varies, $k(n+m-k)$ takes maximum value when $k = \frac{n+m}{2}$. For this reason, we focus on $m = -n, -n+2, \dots, n-2, n$, whereby $k(n+m-k) = (\frac{n+m}{2})^2 - (\frac{n+m}{2} - k)^2$, and therefore $f(x) = f(-2^m) = 2^{(\frac{n+m}{2})^2} \sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2}$. Our

only interest is the sign of this term, and since the sign of $2^{(\frac{n+m}{2})^2} \sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2}$ is

the same as the sign of $\sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2}$, we will focus on the latter.

We isolate the cases $n \leq 2$ first. For $n = 1$ all we have is $x + 1$ so $x = -1$ is a solution, obviously. When $n = 2$ we have $x^2 + 2x + 1 = (x + 1)^2$, so -1 is a double root. Thus we only deal with $n = 3$ here. We recall that if $a_1, a_2, \dots, a_k$ are distinct nonnegative numbers then $\sum_{i=1}^k 2^{-a_i} < \sum_{i=1}^{\infty} 2^{-i} = 1$. Now we have the following cases to consider:

- Case 1: $m = \pm n$. In the $+n$ case we have

$$\sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2} = \sum_{k=0}^n (-1)^k 2^{-(n-k)^2} = (-1)^n + (-1)^{n-1} 2^{-1} + (-1)^{n-2} 2^{-4} + \dots + (-1)^0 2^{-n^2}$$

and by the lemma we had, $|(-1)^{n-1} 2^{-1} + (-1)^{n-2} 2^{-4} + \dots + (-1)^0 2^{-n^2}| \leq 2^{-1} + 2^{-4} + \dots + 2^{-n^2} < 1$ so $(-1)^{n-1} 2^{-1} + (-1)^{n-2} 2^{-4} + \dots + (-1)^0 2^{-n^2} \in (-1, 1)$ which means $\sum_{k=0}^n (-1)^k 2^{-(n-k)^2}$ has the same sign as $(-1)^n$. Similarly, when $m = -n$ the expression

$$\sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2} = \sum_{k=0}^n (-1)^k 2^{-k^2}$$

has the same sign as $(-1)^0 = 1$ (i.e. positive).

- Case 2: now $-n < m < n$ and n has the same parity as n . Then $\sum_{k=0}^n (-1)^k 2^{-(\frac{n+m}{2}-k)^2}$

has the following form:

$$(-1)^0 2^{-(\frac{n+m}{2})^2} + (-1)^1 2^{-(\frac{n+m}{2}-1)^2} + \dots + (-1)^{(n+m)/2} 2^0 + \dots + (-1)^n 2^{-(\frac{n+m}{2}-n)^2}$$

W.L.O.G. assume $m \leq 0$; the other case is symmetric to this. We notice that $(\frac{n+m}{2}-k)^2 = (\frac{n+m}{2} - (n+m-k))^2$, and moreover $n+m$ is even so k and $n+m-k$ has the same parity. This means we can group these terms together for $k = 0, 1, \dots, \frac{n+m}{2}-1$ to get

$$\begin{aligned} & \sum_{i=0}^{\frac{n+m}{2}-1} ((-1)^i + (-1)^{n+m-i}) 2^{-(\frac{n+m}{2}-i)^2} + (-1)^{\frac{n+m}{2}} + \sum_{i=n+m+1}^n (-1)^i 2^{-(\frac{n+m}{2}-i)^2} \\ &= (-1)^{\frac{n+m}{2}} + 2 \sum_{i=0}^{\frac{n+m}{2}-1} (-1)^i 2^{-(\frac{n+m}{2}-i)^2} + \sum_{i=n+m+1}^n (-1)^i 2^{-(\frac{n+m}{2}-i)^2} \\ &= (-1)^{\frac{n+m}{2}} + 2(-1)^{\frac{n+m}{2}-1} 2^{-1} + 2 \sum_{i=0}^{\frac{n+m}{2}-2} (-1)^i 2^{-(\frac{n+m}{2}-i)^2} + \sum_{i=n+m+1}^n (-1)^i 2^{-(\frac{n+m}{2}-i)^2} \\ & \quad 2 \sum_{i=0}^{\frac{n+m}{2}-2} (-1)^i 2^{-(\frac{n+m}{2}-i)^2} + \sum_{i=n+m+1}^n (-1)^i 2^{-(\frac{n+m}{2}-i)^2} \\ & \quad \sum_{i=0}^{\frac{n+m}{2}-2} (-1)^i 2^{-(\frac{n+m}{2}-i)^2+1} + \sum_{i=n+m+1}^n (-1)^i 2^{-(\frac{n+m}{2}-i)^2} \end{aligned}$$

(basically, the two terms beside $(-1)^{(n+m)/2}$ are $(-1)^{(n+m)/2-1} 2^{-1} + (-1)^{(n+m)/2+1} 2^{-1}$ and therefore vanishes). We recognize that the exponents $-(\frac{n+m}{2}-i)^2 + 1$ with $i =$

$0, \dots, \frac{n+m}{2} - 2$ are different numbers in the range $[-(\frac{n+m}{2})^2 + 1, -3]$ and $-(\frac{n+m}{2} - i)^2$ with $i = n+m+1, n$ are different numbers in the range $[-(\frac{m-n}{2})^2, -(-\frac{m+n}{2} - 1)^2]$ and $-(-\frac{m+n}{2} - 1)^2 < -(\frac{n+m}{2})^2 + 1$ are disjoint, which means together all these exponents represent different negative numbers. Therefore by the lemma above, the sign will follow the dominating one, i.e. $(-1)^{(n+m+2)/2}$, i.e. $(-1)^{(n+m)/2}$. This conclusion will hold for $m > 0$ too.

Summarizing above, we know that when $x = 2^m$ for $m = -n, -n+2, \dots, n-2, n$, $f(x)$ follows the sign of $(-1)^{(n+m)/2}$. In particular, $(-1)^{(n+m)/2}$ and $(-1)^{(n+m+2)/2}$ have different signs, so there is a root between $(-2^{m+2}, -2^m)$. Considering $m = -n, \dots, -n+2$ we know that there are roots in the intervals $(-2^n, -2^{n-2}), (-2^{n-2}, \dots, -2^{n-4}), \dots, (-2^{-n+2}, -2^{-n})$ which are n disjoint intervals, hence at least n real roots. On the other hand, f is a polynomial with degree n , hence only at n roots in total. Thus all roots are real.